

Disproof of Conjecture 00000000423: Promotion Orbit Lengths Do Not Always Divide the Tableau Size

TLMC submission

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Abstract

Conjecture 00000000423 asserts that promotion orbit lengths on standard Young tableaux divide the tableau size. We disprove it with the shape $(2, 1)$: the two standard Young tableaux of that shape form a single promotion orbit of length 2, while the tableau size is $n = 3$ cells, and $2 \nmid 3$. The computation is three lines of jeu-de-taquin; it is re-verified by an independent script (`reproduce.py`) and formalized in core Lean 4 with zero axioms (`lean4/`).

1 The conjecture

For a partition λ of n , let $\text{SYT}(\lambda)$ denote the set of standard Young tableaux of shape λ . Schützenberger’s *promotion* operator π acts on $\text{SYT}(\lambda)$ by cyclic sliding: delete the entry 1, perform jeu-de-taquin slides into the vacated cell, subtract 1 from every remaining entry, and place n in the (corner) cell vacated by the slide.

Conjecture 1 (00000000423). *Promotion orbit lengths divide the tableau size; the simultaneous equidistribution and orbit counting are given jointly by a bijection.*

The first clause is the target of this disproof: orbit lengths of π on $\text{SYT}(\lambda)$ should divide the tableau size.

2 The attack: shape $(2, 1)$

Shape $(2, 1)$ has $n = 3$ cells and exactly two standard fillings (entries 1, 2, 3; rows and the first column strictly increasing):

$$A = \begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} \qquad B = \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$$

Claim 1 ($\pi(A) = B$). *Delete the 1 from A ; the empty cell $(1, 1)$ has right neighbour 2 and lower neighbour 3; the smaller, 2, slides left. The vacated cell $(1, 2)$ has no right or lower neighbour, so the slide stops. Subtract 1 from the remaining entries ($3 \mapsto 2$) and place 3 at $(1, 2)$:*

$$A \mapsto \begin{array}{|c|c|} \hline 2 & \cdot \\ \hline 3 & \\ \hline \end{array} \mapsto \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array} = B.$$

Claim 2 ($\pi(B) = A$). *Delete the 1 from B ; the empty cell has right neighbour 3 and lower neighbour 2; the smaller, 2, slides up. The vacated cell $(2, 1)$ has no neighbour, so the slide stops. Subtract 1 ($3 \mapsto 2$) and place 3 at $(2, 1)$, giving A .*

Hence π is the transposition $A \leftrightarrow B$ on $\text{SYT}(2, 1)$: the entire tableau set is a *single orbit of length 2* (and $\pi \neq \text{id}$ since $A \neq B$).

Theorem 1 (Disproof of 00000000423). *π on $\text{SYT}(2, 1)$ has orbit length 2, the tableau size is $n = 3$, and $2 \nmid 3$ ($3 = 2 \cdot 1 + 1$). Therefore the conjecture is false.*

3 Boundary of the attack

- **Rectangles are not counterexamples.** For an $a \times b$ rectangle with $n = ab$ cells, Haiman’s formula gives promotion order $n / \gcd(n, a)$, which divides n . The failure genuinely requires a non-rectangular shape.
- **Minimality.** Under the reading “tableau size = number of cells,” $(2, 1)$ is the smallest counterexample: it is the smallest partition that is not a row, column or rectangle with the divisibility holding.
- **The alternative reading “size = #SYT” also fails.** For shape $(3, 2)$: $n = 5$ cells, $\#SYT = 5$, and the promotion orbits have lengths $\{3, 2\}$ (lcm 6). Neither 3 nor 2 divides 5, and 6 divides neither 3 nor 5. So both natural readings of “tableau size” are refuted, at $(2, 1)$ and $(3, 2)$ respectively.

4 Verification

The tableau enumeration and promotion computation were performed three independent ways:

1. the jeu-de-taquin computation of Claims 1 and 2 above;
2. the script `reproduce.py`, which enumerates SYT of a shape by brute force and computes promotion orbits (shape $(2, 1)$: orbit lengths $[2]$; shape $(3, 2)$: orbit lengths $[2, 3]$, lcm 6);
3. a core Lean 4 formalization (`lean4/`), whose main theorem `conjecture_00000000423_false` states that the tableau set equals $\{(1, 2, 3), (1, 3, 2)\}$, that promotion swaps the two, and that $(3 : \mathbb{N}) \bmod 2 = 1$. `#print axioms` on every theorem reports “does not depend on any axioms”; there are no `sorry`.